

SuperDoc math: m:box / m:borderBox spec coverage test

Each scenario below tests a specific branch of ECMA-376 §22.1.2.11-14 and §22.9.2.7. Labels describe the expected MathML.

A. m:borderBox defaults

1. No borderBoxPr (expect notation=box)

 $\boxed{abc}$

2. Empty borderBoxPr (expect notation=box)

 $\boxed{abc2}$

B. ST_OnOff value variants (on m:hideTop)

3. m:val=1 (expect top hidden)

 $\boxed{T1}$

4. m:val=true (expect top hidden)

 $\boxed{Ttr}$

5. m:val=on (Annex L.6.1.3 uses this — expect top hidden)

 $\boxed{Ton}$

6. Bare tag hideTop/ (expect top hidden — spec default val=1)

 $\boxed{Tbr}$

7. m:val=0 (expect NOT hidden)

 $\boxed{T0}$

8. m:val=false (expect NOT hidden)

 $\boxed{Tf}$

C. Individual hide flags in isolation

9. hideBot only (expect top left right)

 $\boxed{B}$

10. hideLeft only (expect top bottom right)

$\overline{L}$

11. hideRight only (expect top bottom left)

$\overline{R}$

12. hideTop + hideRight — spec §22.1.2.12 example (expect bottom left)

$\underline{x}$

13. All four hides, no strikes (expect fallback to mrow, no border)

$nobdr$

D. Strike directions (all borders hidden)

14. strikeBLTR alone (BL to TR = slash = updiagonalstrike)

$BLTR$

15. strikeTLBR alone (TL to BR = backslash = downdiagonalstrike)

$TLBR$

16. strikeH alone (horizontalstrike)

H

17. strikeV alone (verticalstrike)

V

E. Strike combinations

18. strikeBLTR + strikeTLBR = X pattern

X

19. All four strikes

$\cancel{\cancel{\cancel{\cancel{a}}}}$

20. Default box + strikeH (expect: box horizontalstrike)

$\boxed{boxH}$

21. hideTop + strikeH (expect: bottom left right horizontalstrike)

$\underline{pH}$

F. Spec Annex L.6.1.3 example

22. a squared = b squared + c squared crossed out with TL-BR diagonal, using m:val=on from spec

$$\boxed{a^2 = b^2 + c^2}$$

G. m:box scenarios

23. Plain m:box with content == (expect mrow)

$$==$$

24. m:box with boxPr/opEmu — spec §22.1.2.13 example (opEmu currently dropped)

$$a == b$$

25. m:box with boxPr/noBreak (noBreak currently dropped)

$$nbr$$

26. m:box with all boxPr flags (all currently dropped)

$$pAll$$

27. Empty m:box (expect null/dropped)

H. Nested structures

28. m:box inside m:borderBox

$$\boxed{nx}$$

29. m:borderBox inside m:box

$$\boxed{ny}$$

30. Nested m:borderBox inside m:borderBox (outer partial-hide + inner strike)

$$\boxed{\cancel{nz}}$$